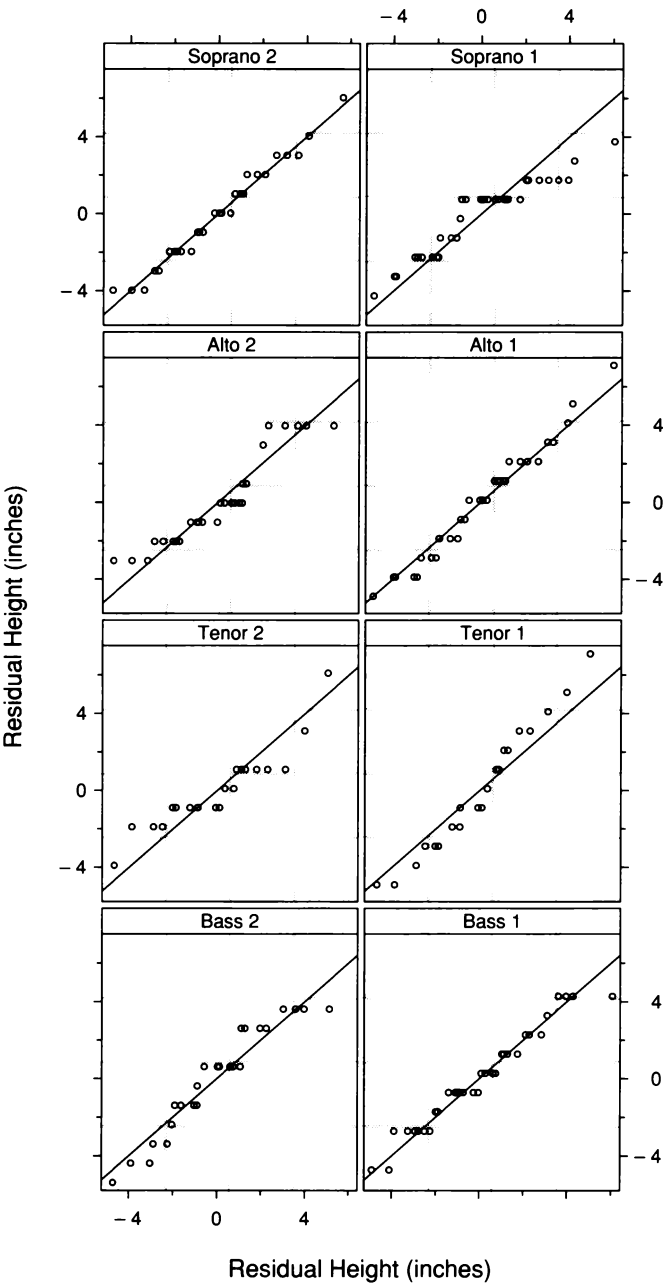


Suppose, as our visualization so far suggests, that the underlying patterns of the eight singer distributions differ by additive shifts. Then the distributions of the eight sets of residuals should be nearly the same because subtracting means eliminates the shifts. Our next task is to compare the distributions of the residuals to see if they appear to be nearly the same; this provides a confirmation of our additive-shift observation.

The residual distributions could be compared by all pairwise q-q plots, but there are 28 pairs, and we would be back to the problem of the pairwise q-q plots of the data in Figure 2.5 — assessing a substantial amount of variation. Figure 2.14 uses another method that results in just eight q-q plots. On each panel, the quantiles of the residuals for one voice part are graphed on the vertical scale against the quantiles of the residuals for all voice parts on the horizontal scale. The line on each panel has slope one and intercept zero. Since the underlying patterns of the points on the eight panels follow these lines, the residual distributions are about the same. This adds credence to a conclusion of additive shifts.

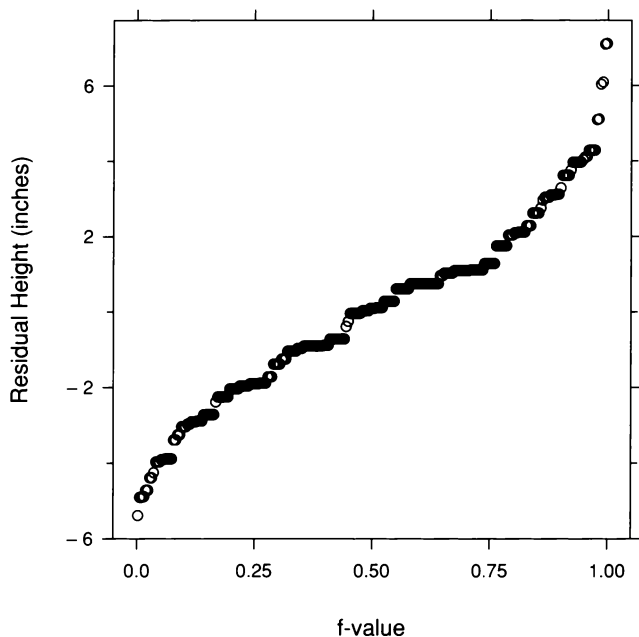
Homogeneity and Pooling

The process of identifying a structure in data and then fitting the structure to produce residuals that have the same distribution lies at the heart of statistical analysis [3, 30, 35, 76]. Such *homogeneous* residuals can be *pooled*, which increases the power of the description of the variation in the data. For the singer data, we have judged the eight residual distributions to be homogeneous, so we can pool them, using the variation in the entire collection of residuals to describe the variation in the residuals for each voice part. This fitting and pooling leads, as we will now show, to a more informative characterization of the variation of the height distribution for each of the voice parts.



2.14 Each panel is a q-q plot that compares the distribution of the residuals for one voice part with the distribution of the pooled residuals.

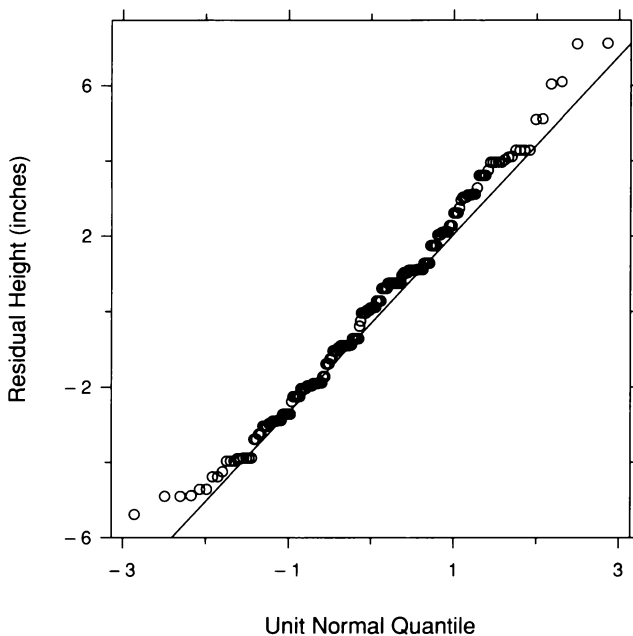
Each voice-part distribution can be described by its mean together with the pooled residual variation. Figure 2.15 is a quantile plot of all of the residuals. The 0.025 quantile of the residuals is -4.6 inches and the 0.975 quantile is 4.8 inches. Thus 95% of the residuals lie between about ± 4.7 inches. Consider the first sopranos. The mean is 64.3 inches. The resulting description of the 95% first soprano variation is 64.3 ± 4.7 inches, which is 59.6 inches to 69.0 inches. Similarly, the second bass variation is 71.4 ± 4.7 inches, which is 66.7 inches to 76.1 inches. We have been able to use the richer information source of all of the residuals to describe the variation in the second basses and the variation in the first sopranos, rather than relying on just the first soprano heights to describe their variation, and just the second bass heights to describe their variation. Of course, the pooling power has come from imposing structure on the data — an additive-shift structure — and the resulting description is valid only if the structure is valid, but the visualization of the data has made the imposition entirely reasonable.



2.15 The pooled residuals are displayed by a quantile plot.

Fitting the Normal Distribution

Once the homogeneity of a set of residuals has been established, we can attempt a fit of the normal distribution to them. Figure 2.16 is a normal quantile graph of the singer residuals. There is a hint of curvature in the underlying pattern, but the effect is relatively minor, so the residual distribution is reasonably well approximated by the normal. Thus we can use a fitted normal to characterize the variability in the residuals.



2.16 A normal q-q plot compares the distribution of the pooled residuals with a normal distribution.

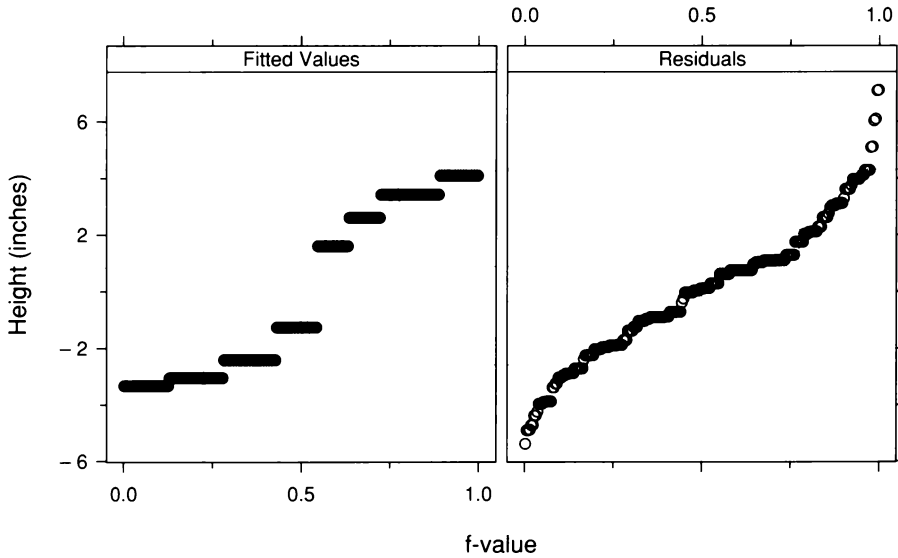
The sample mean of the residuals is 0 inches because the mean of the residuals for each voice part is 0 inches. The sample standard deviation of the residuals is

$$\sqrt{\frac{1}{n-8} \sum_{p=1}^8 \sum_i \hat{\epsilon}_{pi}^2} = 2.45 \text{ inches} .$$

Thus the fitted normal has a mean of 0 inches and has a standard deviation of 2.45 inches. The 95% variation about the mean of a normal distribution is ± 1.96 times the standard deviation; for the singer residuals, this is ± 4.8 inches, a value that is very close to the ± 4.7 inches that arose from the residual quantiles. Thus, using this normal fit, the description of the 95% variability in the first soprano heights is 59.5 inches to 69.1 inches and the 95% variability in the second bass heights is 66.6 inches to 76.2 inches. The description via the normal approximation is attractive because it is more parsimonious than the description based on the residual quantiles.

The Spreads of Fitted Values and Residuals

It is informative to study how influential the voice-part variable is in explaining the variation in the height measurements. The fitted values and the residuals are two sets of values each of which has a distribution. If the spread of the fitted-value distribution is large compared with the spread of the residual distribution, then the voice-part variable is influential. If it is small, the voice-part variable is not as influential. Figure 2.17 graphs the two distributions by quantile plots; since it is the spreads of the distributions that are of interest, the fitted values minus their overall mean are graphed instead of the fitted values themselves. This *residual-fit spread plot*, or *r-f spread plot*, shows that the spreads of the residuals and the fitted values are comparable. Thus the voice-part variable accounts for a significant portion of the variation in the height data, but there is a comparable amount of variation remaining.



2.17 An r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the fit to the singer data.

Fitting and Graphing

This approach to studying the singer distributions — fitting sample means to the data, subtracting them to get residuals, and then graphing the fit and residuals — illustrates the visualization paradigm of fitting and graphing. For the singer heights, it has led us to a more powerful description of the data. The sample means fitted to the data summarize the locations of the distributions. The graphs of the residuals make it clear that the distributions of the residuals are homogeneous and well approximated by the normal distribution. The pooling of the residuals to characterize the variation for each voice part and the approximation of the pooled residual distribution by the normal increase the information in the description of the variation of the data. The final result is a convincing, quantified picture of the relationship between height and pitch interval.

2.6 Log Transformation

In 1960 Bela Julesz sprang an ingenious invention on the scientific world of visual perception — the random dot stereogram [56, 57]. An example, designed by Julesz, is shown in Figure 2.18. Each of the two images has the appearance of a collection of random dots. But when the images are looked at in stereo — which means that one image falls on the left retina and the other image falls on the right — a 3-D object is seen. In this case, the object is a diamond-shaped region floating above or below the plane of the page.



2.18 A diamond-shaped region is shown in 3-D by a random dot stereogram.

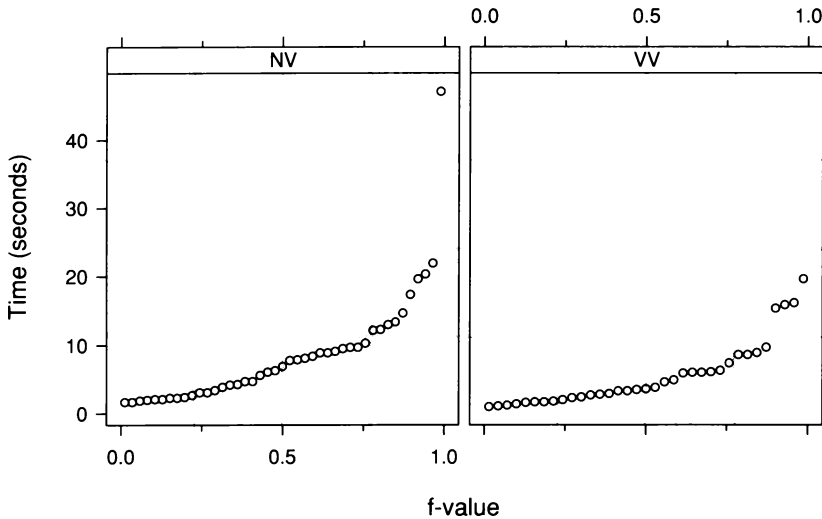
One way to fuse the two images in Figure 2.18 — that is, to see the 3-D effect — is to use a stereo viewer, which channels the left image to the left retina and the right image to the right retina. This makes the diamond float above the page. Another method requires no equipment — fixing on a point between the two images and defocusing the eyes by looking through the images, bringing them together in the middle. It is helpful to vary the distance of the image from the eyes to find an optimal position for fusion. When fusion is accomplished, the visual system actually sees three images, with the center one in 3-D, and, again, the diamond floats above the page. It is hard to make this method work the first time, but after a first success, it is typically easy to do again. A third method is crossing the eyes; since this sends the left image to the right retina and the right image to the left retina, the diamond floats below the page. For those without a stereo viewer, mastering the technique of viewing with the unaided eye will have a benefit. A visualization method in Chapter 4 uses stereo to display trivariate data.

The idea of the random dot stereogram is simple and elegant. Embedded in the right view is a region, in this case a diamond, that is exactly repeated in the left view, but is shifted slightly to the right. Quite incredibly, this *binocular disparity* is enough to enable fusion. Julesz's invention demonstrated that the visual system can process local detail to construct an object in 3-D, and needs no other information about the form of the object.

Typically, a viewer concentrating on a random dot stereogram achieves fusion after a few seconds or more. The fusion is not achieved by conscious thought, for example, by the processes that allow us to reason about the world. But fusing the stereogram once makes it easier to do again, so something stored in the brain's memory can decrease fusion time. An experiment was run to study the effect of prior knowledge of an object's form on fusion time [44]. The experimenters measured the time of first fusion for a particular random dot stereogram. There were two groups of subjects. The NV subjects received either no information or verbal information. The VV subjects received a combination of verbal and visual information, either suggestive drawings of the object or a model of it. Thus the VV subjects actually saw something that depicted the object, but the NV subjects did not. The goal in analyzing the fusion times is to determine if there is a shift in the distribution of the VV times toward lower values compared with the NV times. The experimenters used a classical method of probabilistic inference to analyze the data, and concluded that there is no shift. We will use visualization methods to re-examine this result.

Skewness

Figure 2.19 shows quantile plots of the fusion times. The data are *skewed toward large values*. Small values are tightly packed together, and the large values stretch out and cover a much wider range of the measurement scale. The extreme case is the largest NV time, 47.2 seconds, which is more than twice the value of the next largest observation. The skewing gradually increases as we go from small to large values; the result is a strongly convex pattern.

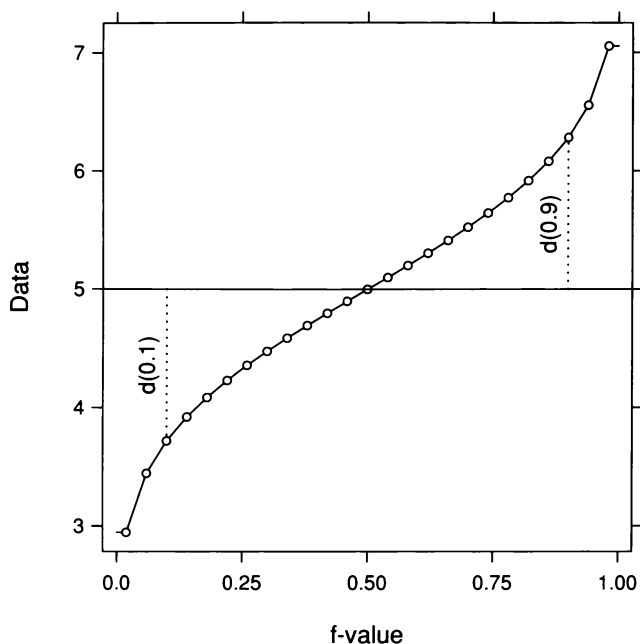


2.19 Quantile plots display the two distributions of the fusion-time data.

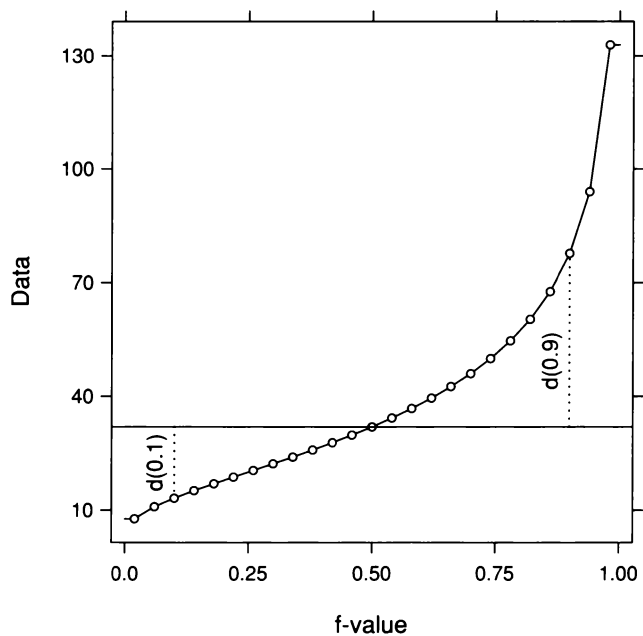
As before, let $q(f)$ be the f quantile of the data. Let $d(f)$ be the distance of $q(f)$ from the median,

$$d(f) = |q(0.5) - q(f)|.$$

A distribution is symmetric if $d(f)$ is symmetric about 0.5 as a function of f ; that is, $d(f) = d(1 - f)$. The values shown on the quantile plot in Figure 2.20 are symmetric. A distribution is skewed toward large values if $d(f)$ is bigger than $d(1 - f)$ for f in the interval 0.5 to 1, and the disparity increases as f goes from 0.5 to 1, that is, from the center of the distribution to the tails. The values shown on the quantile plot of Figure 2.21 are skewed toward large values.

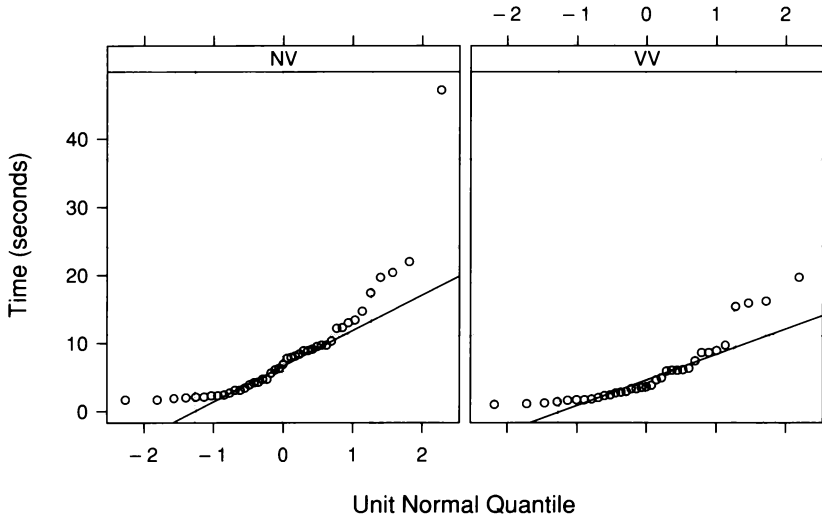


2.20 The graph shows the quantile function of data with a symmetric distribution.



2.21 The graph shows the quantile function of data that are skewed toward large values.

Figure 2.22 shows normal q-q plots of the fusion times. The skewness toward large values creates the same convex pattern as on the quantile plot. This deviation from normality occurs because the normal distribution is symmetric. This behavior contrasts with that of the distributions of the singer heights, which are nearly symmetric and are well approximated by the normal distribution.

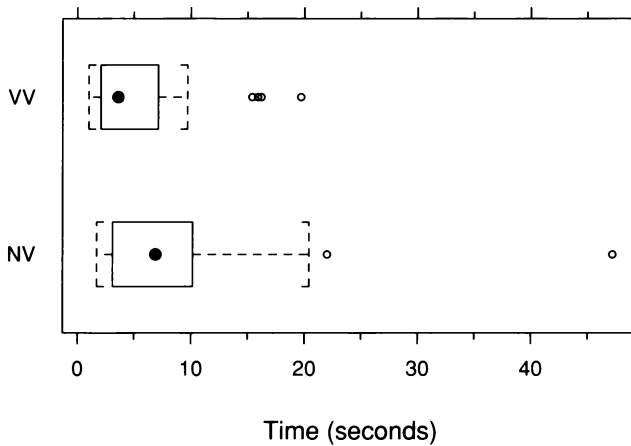


2.22 Normal q-q plots compare the distributions of the fusion-time data with the normal distribution.

Data that are skewed toward large values occur commonly. Any set of positive measurements is a candidate. Nature just works like that. In fact, if data consisting of positive numbers range over several powers of ten, it is almost a guarantee that they will be skewed. Skewness creates many problems. There are visualization problems. A large fraction of the data are squashed into small regions of graphs, and visual assessment of the data degrades. There are characterization problems. Skewed distributions tend to be more complicated than symmetric ones; for example, there is no unique notion of location and the median and mean measure different aspects of the distribution. There are problems in carrying out probabilistic methods. The distribution of skewed data is not well approximated by the normal, so the many probabilistic methods based on an assumption of a normal distribution cannot be applied. Fortunately, remedies coming in later sections can cure skewness.

Monotone Spread

In Figure 2.23, box plots of the fusion-time distributions show that the median of the NV times is greater than the median of the VV times, and the spread of the NV times is greater than the spread of the VV times. In other words, the spreads increase with the locations. The same phenomenon can also be seen on the normal q-q plots in Figure 2.22. The slope of the line on each panel is the interquartile range of the data divided by the interquartile range of the unit normal distribution. This measure of spread is just a rescaling of the interquartile range of the data. In Figure 2.22, the slope for the NV times is greater than the slope for the VV times.



2.23 The distributions of the fusion times are compared by box plots.

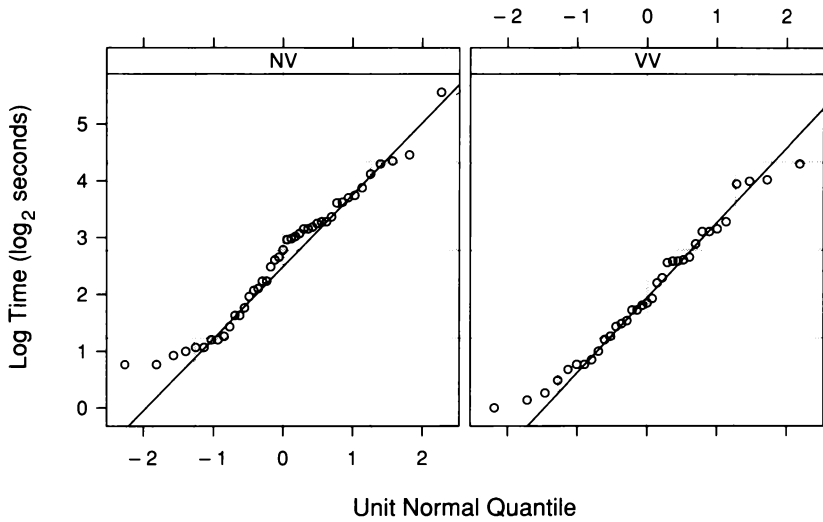
When the distributions of two or more groups of univariate data are skewed, it is common to have the spread increase monotonically with location. This behavior is *monotone spread*. Strictly speaking, monotone spread includes the case where the spread decreases monotonically with location, but such a decrease is much less common for raw data. Monotone spread, as with skewness, adds to the difficulty of data analysis. For example, it means that we cannot fit just location estimates to produce homogeneous residuals; we must fit spread estimates as well. Furthermore, the distributions cannot be compared by a number of standard methods of probabilistic inference that are based on an assumption of equal spreads; the standard t-test is one example. Fortunately, remedies for skewness can cure monotone spread as well.

Transformation by Logs

For positive univariate measurements, it is often more natural to consider multiplicative effects rather than additive ones. The fusion times are one example. Consider times of 4 seconds and 6 seconds for two individuals. The interesting information is that the second individual took 1.5 times as long. The two individuals differ by a lot more in their fusion performance than two individuals with times of 20 seconds and 22 seconds. Even though the absolute difference, 2 seconds, is the same in both cases, the poorer performer in the first case took 50% longer, whereas the poorer performer in the second set took only 10% longer. Taking logs amounts to changing the units of the data — from seconds to log seconds for the fusion times — in such a way that equal differences now mean equal multiplicative factors. This simplifies the interpretation of the measurement scale because, to put it simplistically, addition is easier than multiplication.

The logarithm is one of many transformations that we can apply to univariate measurements. The square root is another. Transformation is a critical tool for visualization or for any other mode of data analysis because it can substantially simplify the structure of a set of data. For example, transformation can remove skewness toward large values, and it can remove monotone increasing spread. And often, it is the logarithm that achieves this removal.

Figure 2.24 shows normal q-q plots of the logs of the fusion times. The data are now much closer to symmetry, although there is a small amount of remaining skewness; the points in the lower tail of each distribution lie somewhat above the line. In other words, the data quantiles in the lower tail are a bit too big, or too close to the median. The effect is small, but there is a plausible explanation. There is a minimum amount of time that a subject needs to consciously realize that the visual system has fused the stereogram, and then to signal that fusion has occurred. Effectively, this makes the origin somewhat bigger than 0 seconds. We could try to bring the lower tail into line by subtracting a small constant such as 0.5 seconds from all of the data, but this gets fussier than we need to be.



2.24 Normal q-q plots compare the distributions of the log fusion times with the normal distribution.

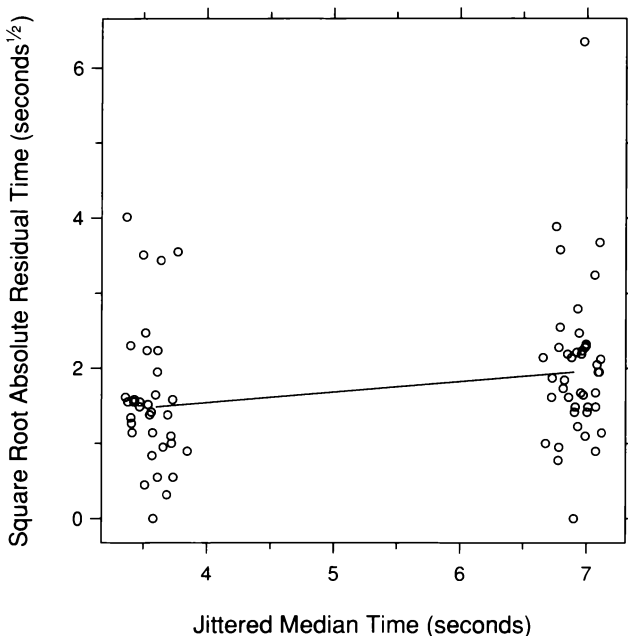
In graphing the log fusion times, log base 2 has been used. Thus, a change of one unit on the transformed scale means a change on the original scale by a factor of two. When data range through just a few powers of 10, $\log_2$ is easier to interpret than $\log_{10}$ because fractional powers of 10 are harder to fathom than integer powers of 2 [20].

Except for the outlier, the fusion times range from 1 second to 22 seconds. On the $\log_2$ scale, this range becomes 0 $\log_2$ seconds to 4.5 $\log_2$ seconds. On the $\log_{10}$ scale, the range becomes 0 $\log_{10}$ seconds to 1.3 $\log_{10}$ seconds, and we must expend effort fathoming fractional powers of 10 to comprehend the multiplicative effects.

S-L Plots

Monotone spread, when it occurs, can typically be spotted on box plots. For example, Figure 2.23 revealed the monotone spread of the fusion times. But the box plot is a general-purpose visualization tool for exploring many aspects of distributions, not just spread. The *spread-location plot*, or *s-l plot*, provides a more sensitive look at monotone spread because it is a specialized tool whose sole purpose is to detect changes in spread. The s-l plot translates looking at spread to looking at location. First, medians are fitted to the distributions of the data, x_i . Measures of location for the absolute values of the residuals, $|\hat{\varepsilon}_i|$, are measures of spread for the x_i . For example, the *median absolute deviations*, or *mads*, of the distributions of the x_i are the medians of the distributions of the $|\hat{\varepsilon}_i|$ [51, 65].

Figure 2.25 is an s-l plot of the fusion times. The circles graph the square roots of the $|\hat{\varepsilon}_i|$ for the two distributions against the fitted values, which take on two values, the medians of the two distributions. To prevent undue overlap of the plotting symbols, the locations of the symbols are jittered by adding uniform random noise to the fitted values. Jittering will be discussed further in Chapter 3.

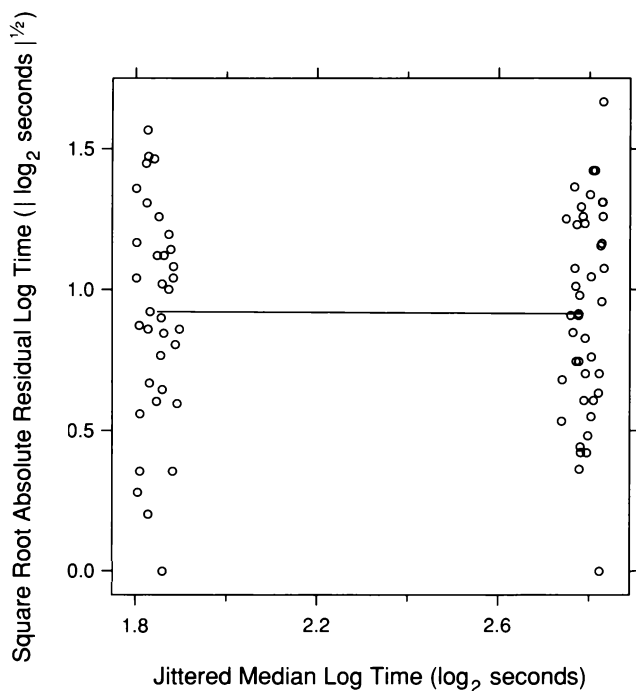


2.25 The s-l plot for the fusion times checks for nonuniform spread.

The square root transformation is used because absolute residuals are almost always severely skewed toward large values, and the square root often removes the asymmetry. Also, the square roots of the two mads are graphed against the two medians by connecting the plotting locations by a line segment.

The s-l plot for the fusion times adds to our understanding of the monotone spread of the distributions. It shows a convincing upward shift in the location of the $\sqrt{|\hat{\varepsilon}_i|}$ for the NV times.

Figure 2.26 is an s-l plot for the log fusion times. The log transformation, in addition to removing most of the skewness, makes the spreads nearly equal. It is not unusual for a single transformation to do both. Nature is frequently kind enough to allow this.

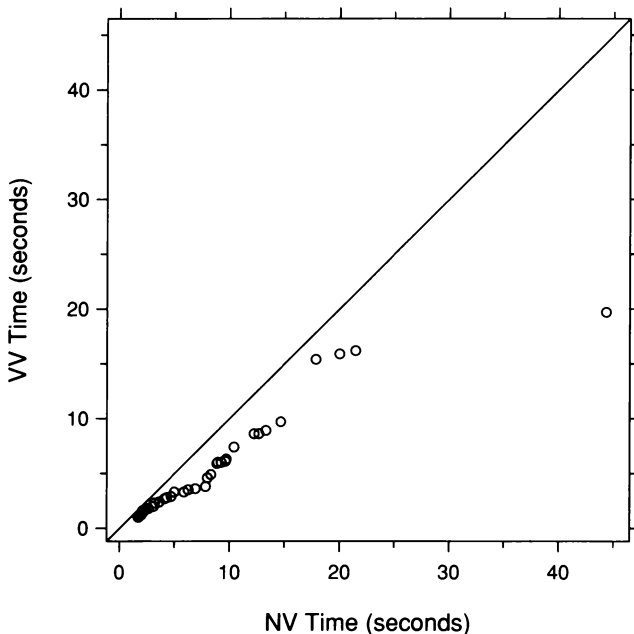


2.26 The s-l plot for the log fusion times checks for nonuniform spread.

Multiplicative and Additive Shifts

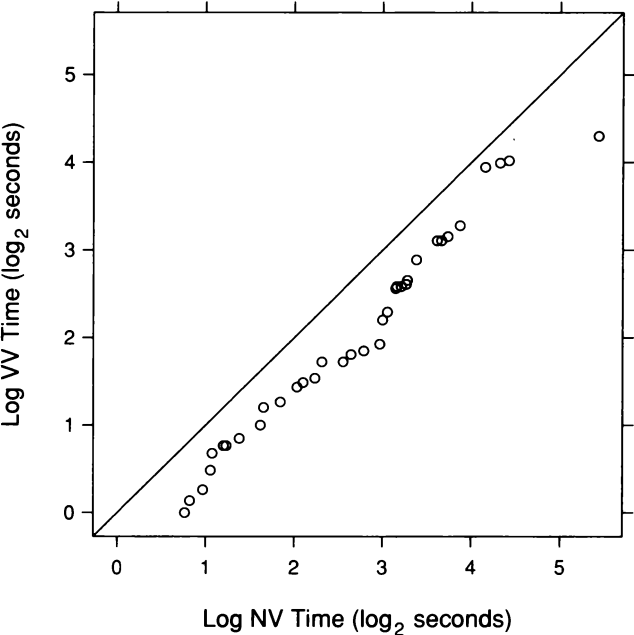
As we saw for the fusion times, logs can simplify structure by changing severe skewness to near symmetry and by changing monotone spread to nearly equal spread. It can also simplify structure by changing multiplicative shifts among distributions to additive shifts.

Figure 2.27 is a q-q plot of the fusion times on the original scale, before taking logs. There is a shift in the two distributions, but unlike the singer heights, it is not additive. The underlying pattern is a line through the origin with a slope of about $2/3$. The shift is multiplicative, and quantiles with large f -values differ by more than those with small ones. This multiplicative shift is more complicated than an additive one because it results in a difference not just in the locations of the distributions but in the spreads as well. This produces the monotone spread.

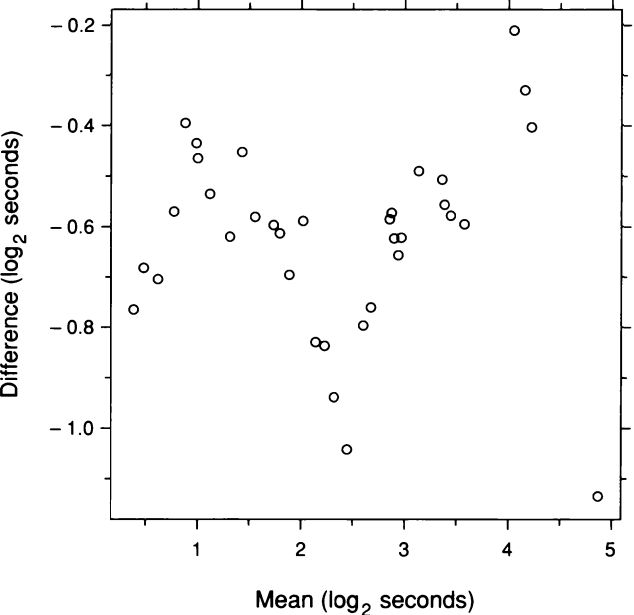


2.27 The q-q plot compares the distributions of the fusion times.

Figure 2.28 is a q-q plot of the log fusion times, and Figure 2.29 is a corresponding m-d plot. Now, on the log scale, the effect is additive. On the average, the log VV times are about $0.6 \log_2$ seconds less than the log NV times. Back on the original scale, this is just the multiplicative effect with a multiplicative constant of $2^{-0.6} = 0.660 \approx 2/3$.



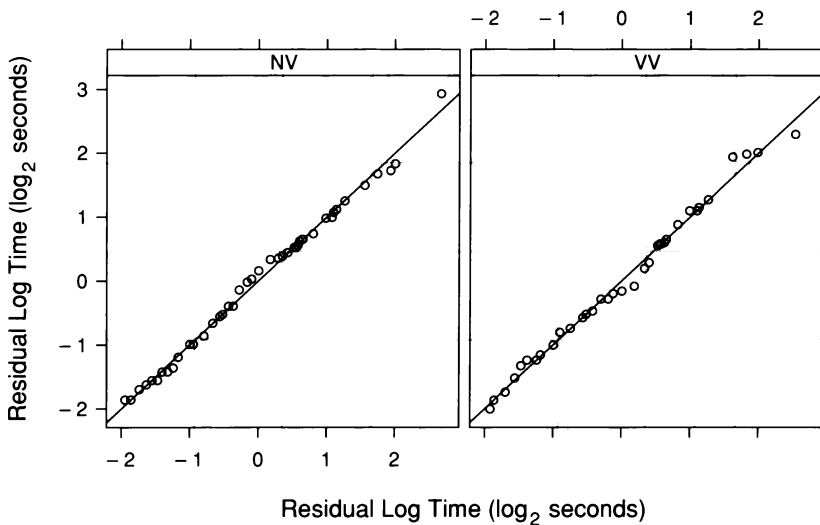
2.28 The q-q plot compares the distributions of the log fusion times.



2.29 The m-d plot provides more information about the shift of the log fusion times.

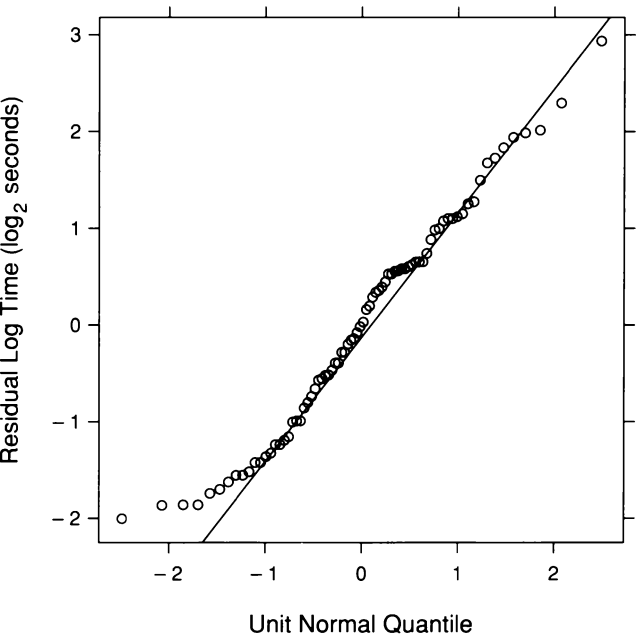
Fitting and Residuals

Since logs result in an additive shift of the fusion times, the fitting on the log scale only needs to account for a shift in location. Since the distributions are not far from the normal, means can be used to estimate locations. The mean for the log NV times is $2.6 \log_2$ seconds, and the mean for the log VV times is $2.0 \log_2$ seconds. Figure 2.30 shows q-q plots of the residuals; the quantiles of each residual distribution are graphed against the quantiles of the pooled residuals. The patterns lie close to the lines, which have intercept 0 and slope 1, so the two residual distributions are nearly the same. The residuals are homogeneous and can be pooled to characterize the variation in the data.

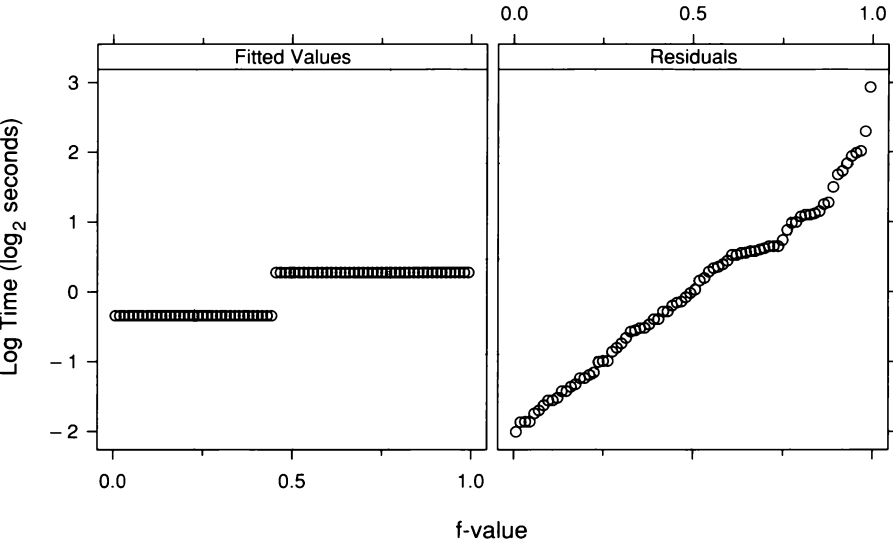


2.30 Each panel is a q-q plot that compares the distribution of the residuals for one group of log times with the distribution of the pooled residuals.

Figure 2.31, a normal q-q plot of the pooled residuals, shows the lifting of the lower tail observed earlier. The departure from normality begins at a value of -1 on the horizontal axis. This is the 0.16 quantile of the unit normal distribution. Thus the upper 84% of the distribution of the pooled residuals is well approximated by the normal. Figure 2.32, an r-f spread plot, shows that the spread of the residuals is considerably greater than the spread of the fitted values. Thus the effect of the increased VV information is small compared with other factors that affect fusion time.



2.31 A normal q-q plot compares the normal distribution with the distribution of the pooled residuals for the fit to log fusion time.



2.32 An r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the fit to log fusion time.